



Cadence IPs for Automotive Ethernet MAC

Part Number: IP701xA

IP Rev: r1p12 - Doc Rev: 1.1

Safety Manual

September 22, 2017

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About this Document

This Safety Manual provides an overview of the safety features and requirements of Cadence IPs for Automotive Ethernet MAC (as listed in *Table 1*) in safety-related automotive designs according to ISO 26262. The intended audience of this document is safety system developers.

This manual is organized as follows:

Section 1 Introduction on page 7

Section 2 Safety Requirements on page 11

Section 3 Internal Safety Mechanism on page 15

Section 4 External Safety Mechanism on page 22

Part Numbers

This manual is applicable to the Cadence product part numbers listed in the following table:

Table 1: Part Numbers

Part Numbers	Cadence IP for
IP7017A	2.5G Ethernet MAC with DMA, 1588, TSN/AVB, and PCS for automotive
IP7014A	Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB for automotive

References and Related Documents

This manual references the following documents:

Table 2: References and Related documents

Reference No.	Title and Document Number
[01]	Cadence IP for 2.5G and Gigabit Ethernet MAC User Guides, 2017

Acronyms and Abbreviations

The following acronyms and abbreviations are used in this manual:

Table 3: Acronyms and Abbreviations

Term	Definition
AHB	AMBA® AHB Interface
ASF	Advanced Safety Features
ASIL	Automotive Safety Integrity Level
AVB	Audio-Video Bridging
AXI	AMBA AXI Interface
BOM	Bill of Material
Byte	An 8-bit data item
CCF	Common Cause Failure
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
DC	Diagnostic Coverage
DMA	Direct Memory Access
DPRAM	Dual-Port RAM
ECC	Error Correcting Code

EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
FEC	Forward Error Correction
FIT	Failures In Time
FM	Failure Mode
FMEDA	Failure Modes, Effects, and Diagnostic Analysis
FTT	Fault Transition Time
GEM	Gigabit Ethernet MAC
GMI	Gigabit Media Independent Interface
GUI	Graphical User Interface
HAZOP	HAZard and Operability analysis
HW	Hardware
IO	Input / Output
IP	Internet Protocol
LFM	Latent Fault Metric
LPI	Low Power Idle
LSO	Large Send Offload
MAC	Media Access Controller
MDC	Management Data Clock
MDIO	Management Data Input Output
PCS	Physical Coding Sublayer
PFC	Priority Flow Control
PMHF	Probabilistic Metric for random Hardware Failures
PTP	Precision Time Protocol
QoS	Quality of Service
RAM	Random Access Memory
ROM	Read Only Memory
RSC	Receive Side Coalescing
RX	Receiver
SECCED	Single Error Correction and Double Error Detection
SEooC	Safety Element out of Context
SG	Safety Goal
SoC	System-on-Chip
SPF	Single Point Fault
SPFM	Single Point Fault Metric
SPI	Serial Packet Interface
SRAM	Static Random Access Memory
SW	Software
TCP	Transmission Control Protocol
TSN	Time-Sensitive Networking
TSU	Timestamp Unit
TX	Transmitter
UDP	User Datagram Protocol
UFO	UDP Fragmentation Offload

1 Introduction

1.1 Scope

This document is intended to support safety system developers in building their safety-related systems using the built-in mechanisms of Cadence IPs for Automotive Ethernet MAC (as listed in *Table 1*), subsequently also referred to in this manual as the Controller IP. It describes any additional system-level hardware or software safety mechanisms that must be implemented to help achieve the desired system Safety Integrity Level.

1.2 Safety Manual Assumptions

During the evaluation of the Controller IP for safety-related applications, assumptions are made on the system-level environment in which the Controller IP is used. The safety system developer must confirm the validity of any assumptions stated in this safety manual.

1.3 Functional Safety Standards

It is assumed that the safety system developer is familiar with the ISO 26262 functional safety standard. As the Controller IP is a component developed separately from the end application in which it may be used in, the Controller IP is considered to be a Safety Element out of Context (SEooC) development.

1.4 Design Configuration

The Controller IP is highly configurable to suit a multitude of different end applications from simple Ethernet controller with basic functionality for home and mobile use to advanced features used in high end server applications. This configuration is performed through Verilog define options within the delivered example defines files which the customer may tailor to suit the final application. A simple GUI is also provided to aid the customer in setting up the design options.

Some of these configuration options results in significant changes to the area and interfaces of the design for which the FMEDA is based on. A representative design configuration was chosen for running the design through the FMEDA process. This was defined as follows:

1. The line side interface was configured to be GMII only.
2. For the MAC, 4 specific address filters were included.
3. For RX packet inspection and queue assignment, 4 Type1 Screeners, 4 Type2 Screeners, 4 ether-type match registers and 12 compare match registers were included.
4. The Priority Flow Control (802.1Qbb, or PFC) feature was not included.
5. Statistics and snapshot registers are not included. These are considered informational only and not safety critical but would have a sizeable impact on the design area.
6. A timestamp unit (TSU) is included for 1588 support.
7. Support for 802.1Qav and 802.1Qbv is included.
8. Support for 802.3br is not included, addition of this option will significantly increase the size of the design and is not expected to be used in Automotive applications.
9. Support for 802.1CB frame elimination is not included. This feature requires the target system to have additional compatible hardware.
10. The host interface was AXI with a 64-bit data bus width and 32-bit address bus width.
11. Support for up to 3 priority queues.
12. The DMA descriptor buffer depths were set to 2. It is not expected large descriptor buffers will be needed for automotive applications.
13. The maximum number of AXI outstanding transactions was configured to be 16.
14. The TX SRAM used was dual port 16KB, each location being 64-bits data with an 11bit address. The actual width of the SRAM requires an additional 10-bits for the address and ECC protection bits.
15. The RX SRAM used was dual port 16KB, each location being 64-bits data with an 11bit address. The actual width of the SRAM requires an additional 10-bits for the address and ECC protection bits.
16. IP/TCP offload engines were not included. This includes the features for LSO (large send offload), UFO (UDP fragmentation offload) and RSC (receive side coalescing).

17. Partial Store and Forward (cut-thru) mode of operation for the DMA was not included. I.e. it is assumed a full store and forward approach will always be used for automotive (which will eliminate the possibility of TX underflows and removes the need for software to handle frame recovery).
18. The following optional Active Safety Features are included as part of the `gem_asf_enable` Verilog define:
19. Control/Configuration and Status register protection.
20. Data and Address Path parity protection.
21. ECC protection is added to the SRAM data interfaces. This adds Single Error Correction and Double Error Detection (SEDED) capability.
22. TSU duplication protection.
23. Transmit scheduler duplication protection.
24. Host side parity interfaces for AXI address and data buses.

Note that significant changes to the design configuration will mean the SPFM/LFM calculation for the IP that was calculated as part of the FMEDA process will no longer apply and a new calculation based on the changed design will be required. While Cadence can provide estimates of what the effect would be, an audited FMEDA would be required if a new certificate is required. It is recommended that wherever possible, the customer should use the same representative design configuration in order to avoid that step.

2 Safety Requirements

2.1 Safety Goals

The Controller IP is used for transport of Ethernet frames across a physical medium. The information carried may be safety related depending on the usage application of the IP, e.g. reverse parking camera, driver assist sensors etc. The following are the high-level requirements of the IP which if fails, may violate the safety goal of the system:

- DMA descriptor address corruption
- DMA descriptor write data corruption
- DMA data buffer address corruption
- DMA data buffer write data corruption
- No data on outgoing AXI port (RX lockup)
- Status read data corruption via APB
- Interrupt output corruption
- TX SRAM output corruption
- RX SRAM output corruption
- Outgoing Ethernet packet corruption
- Outgoing Ethernet packet stream bandwidth allocation corruption (QoS)
- No data on outgoing Ethernet port (TX lockup)
- TSU timer corruption
- MDIO access corruption (causing failure to configure PHY)

The above safety goals reflect the overall functions of the IP and does not necessarily map directly to the safety goals of the overall system in which the IP is employed in.

2.2 Automotive Safety Integrity Level

The Controller IP is designed to be used in safety relevant applications. The end application of the IP is unknown at the time of development so it is considered to be a SEooC development as defined within the ISO 26262 standard. This allows the IP to be certified independently of the end application.

Using the safety goals defined in *section 2.1*, they are mapped into an FMEDA template provided by SGS-TÜV which is an accredited testing authority. The IP is first broken down into sub-components that are then analyzed to determine possible fault conditions which can subsequently violate one or more safety goals.

This process was completed in conjunction with SGS-TÜV and the Controller IP achieved an ASIL-B Ready classification based on implementation of diagnostic mechanisms defined in *section 4*.

As of revision r1p11 of the Controller IP, a number of internal safety mechanisms have been incorporated to reduce the number of external diagnostic mechanisms that needs to be implemented. These new safety mechanisms are described in *section 3*.

2.3 Fault Tolerance Time

Given the application-dependent nature of the system in which the Controller IP may be employed, the fault detection and recovery time may depend on many system dependent factors such as configured operating speed (10/100/1000Mbps), processor speed and system software reaction time. As an example, when operating at 10Mbps mode the time to transmit a 1522-byte frame on the link side interface is over 1.2ms whereas in 1Gbps mode this time reduces significantly to 12μs. For implementation of the 'lockup detect' external safety mechanisms defined in *sections 3 and 4*, the system software should be aware of the configured operating speed to determine the correct time value to define as a failure condition.

2.4 Base Failure Rate

This is based on IEC TR62380 (2004-08).

MATHEMATICAL MODEL :

$$\lambda = \underbrace{\left\{ \lambda_1 \times N \times e^{-0.35 \times a} + \lambda_2 \right\} \times \left[\frac{\sum_{i=1}^y (\pi_i)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right]}_{\lambda_{die}} + \underbrace{\left\{ 2.75 \times 10^{-3} \times \pi_{\alpha} \times \left(\sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68} \right) \times \lambda_3 \right\}}_{\lambda_{package}} + \underbrace{\left\{ \pi_I \times \lambda_{EOS} \right\}}_{\lambda_{overstress}} \times 10^{-9} / h$$

Since the design is IP and no technology-related information is known, only the first part of this equation is covered by our FMEDA.

$$\lambda_{Die_IP} = (\lambda_1 \times N \times e^{-0.35 \times a} + \lambda_2) \times 10^{-9} / h$$

Where:

- 'N' is the number of transistors in the IP
- 'λ₁' is the per transistor base failure rate of the integrated circuit
- For IP this is 3.4 x 10⁻⁶
- 'a' is the year of manufacturing – 1998
- 2017-1998 = 19
- 'λ₂' is the failure rate related to the technology mastering of the IC and is assumed to be 1.7 (This should be defined by the SoC integrator)

For a typical representative configuration of the IP synthesized in a TSMC 28HPM process and an AXI frequency of 250MHz, N was calculated as:

- IP Gate Count (NAND2 equivalent) = 337.9K
- Number of transistors per gate = 4 (assumption)
- N = 1.35M

In addition, the following assumptions were made:

- Total transistor count of SoC: 200 Million.
- IP as a percentage of overall chip: 0.68%

The base failure rate based on these values was calculated at 0.0174 FIT.

2.5 SPFM and LFM Results

The following are snapshots of the results from the FMEDA calculation for SPFM and LFM with all internal safety mechanisms defined in *section 3* configured and enabled.

	SPFM	LFM
Overall IP / Chip Metrics	90.734%	97.913%
Overall IP / Chip Metrics for transient faults	90.708%	98.084%
Overall IP / Chip Metrics for permanent and transient faults	90.709%	98.083%

Note that for the above calculations, the impact of the SRAMs required by the IP are not factored in as they are external components not instantiated within the IP. However as defined in *section 3*, the IP implements SECDED ECC protection for the SRAM which will greatly affect the results as shown below:

	SPFM	LFM
Overall IP / Chip Metrics	98.790%	99.951%
Overall IP / Chip Metrics for transient faults	98.789%	99.955%
Overall IP / Chip Metrics for permanent and transient faults	98.789%	99.955%

This was based on the following assumptions:

- Memory types set to 'Double access SRAM'
- Total number of memory cells = $2 \times 2048 \times 74 = 303104$
- Assuming an average of 8 transistors per bit
- Percentage of chip for the SRAM is expressed as assumption of 8Mbits of memory on chip

2.6 PMHF Rough Calculation

The following are snapshots of the results from the FMEDA rough calculations for Probabilistic Metric for random Hardware Failures (PMHF):

PMHF rough calculation (worst case with $\beta = 0.47$)	$\Sigma\lambda_{MPF}$ [FIT]	β -factor	PMHF Result [FIT]
$PMHF = \Sigma\lambda_{SPF} + \Sigma\lambda_{RF} + \beta * \Sigma\lambda_{MPF\ latent}$	0.0003	0.47	0.0018

PMHF rough calculation incl. transient faults (<i>not ISO 26262 related</i>)	$\Sigma\lambda_{MPF}$ [FIT]	β -factor	"PMHF $\downarrow$ " Result [FIT]
$PMHF = \Sigma\lambda_{SPF} + \Sigma\lambda_{RF} + \beta * \Sigma\lambda_{MPF\ latent} + \Sigma\lambda_{SPF_t} + \Sigma\lambda_{RF_t} + \beta * \Sigma\lambda_{MPF_t}$	0.0000	0.47	0.1762
	$\Sigma\lambda_{MPF_t}$ [FIT]	β -factor $\downarrow$	
	0.0303	0.47	

2.7 Normal Operation

The normal operation of the Controller IP is defined as the IP operating in accordance to specifications as defined in the Controller IP *User Guide [Table 1]* with no faults indicated through either the internal or externally implemented diagnostic mechanisms.

2.8 Failure Behavior

If a fault occurs, the behavior of the Controller IP and its effect on the overall safety system may be undefined. Therefore, on detection of such faults one of the defined safe states should be employed.

2.9 Safe States

The IP safe states following a fault condition are defined as one of the following:

- Disabled
- Held in reset
- Powered off

In these states the Controller IP interfaces are put into a quiescent state to avoid propagating any faults to the wider system.

2.9.1 Safe State Assumptions

Within the context of functional safety, it is assumed that the defined safe states where the Controller IP is not operational (i.e. disabled) does not violate the functional safety requirements of the system that the Controller IP is used in.

Within the context of functional safety, it is assumed that the defined safe states where the Controller IP is not operational (i.e. disabled) does not violate the functional safety requirements of the system that the Controller IP is used in. If the failure of the Controller IP adversely affects the safety of the system, then the overall system must be transitioned into a safe state.

2.9.2 Transition into Safe State

It is expected that the Controller IP (and possibly overall system) is transitioned into one of the defined safe states if the IP indicates a detectable exception condition has occurred or if one of the defined external safety mechanisms have detected possible abnormal operation.

The IP may be disabled by software clearing the *enable_transmit* and *enable_receive* fields in the *network_control* register located at address 0x000.

If the IP is disabled through software writing to the *network_control* register, then it is recommended that all IP configuration and control registers are re-initialized to ensure they are confirmed to be in a known state.

Where possible, a hardware reset (by activating all IP reset input pins) is recommended.

The software controlling the IP should re-initialize all relevant operations required to bring the IP back into normal operation to transition out of the safe state.

2.10 Recovery from a Fault Condition

In the event of a fault occurring, the Controller IP should be transitioned into a safe state defined in *section 2.9*. It is possible that this fault is due to a transient fault which would be recoverable, therefore the system integrator should consider implementing some form of diagnostic mechanism such as a basic packet transfer test as defined in *section 4.2* to validate the correct functioning of the Controller IP.

If the fault is transient, then this follow up diagnostic should complete without any additional fault condition being reported and the Controller IP may continue to be used as normal. It is the responsibility of the system software to re-initialize the Controller IP software and retry transmission of any pending packets.

If the fault is due to a permanent component failure, the follow up diagnostic tests should result in another fault being reported. In this case, the Controller IP should not be relied on.

2.11 Clock and Power Safety Considerations

Clocks are considered to be top-level inputs to this IP and failure of these input clocks are considered to be the responsibility of the IP Integrators FMEDA analysis and is out of scope of this IP-level FMEDA & Safety Manual.

Similarly, power to this soft IP is considered to be out of scope of this IP-level FMEDA & Safety Manual. Supply of power and associated failure modes are considered to be the responsibility of the IP Integrator.

3 Internal Safety Mechanism

The Controller IP contains internal checks for various exception events which may be configured to generate an interrupt when any of the following exceptions occurs:

- AMBA host bus response errors
- Receive overrun
- Transmit retry exceptions
- Transmit underrun (should not be possible to occur in this application case as design is configured for full store and forward mode)

In addition, the following internal safety mechanisms are available from r1p11 of the IP and should be enabled for continuous monitoring to protect against anomalous behavior:

- Transmit data path lockup detection
- Receive data path lockup detection
- Configuration, control and status registers corruption detection
- Data and address path parity protection
- TSU timer corruption detection
- Transmit scheduler corruption detection

The fault logging and reporting feature of the IP also allows the user to optionally configure interrupt generation for several other internally detected protocol errors which are normally informational only but may be of interest in a safety critical application. These are described in *section 3.12*.

Some guidance is provided in the following sections on how to handle fault notifications from each of these mechanisms.

Note that the internal safety mechanisms are designed to be able to detect permanent as well as transient faults.

3.1 AMBA Host Bus Response Errors

The host system may respond with an error if the Controller IP issues an invalid AMBA request. This may be due to some internal corruption of the DMA address generation resulting in an access to an illegal address etc. This may also be caused by faulty system software programming illegal addresses into either the MAC descriptor configuration or in the descriptor buffers in system memory.

When this error occurs, the Controller IP will generate an interrupt (if not masked) to the software so that the system may be re-initialized as the system may be in an indeterminate state.

The system integrator should determine the system behavior in the event of a host bus error occurring.

This error is reported via bit 11 of the generic interrupt status register (separate for each queue) and in bits 18 and 19 of the ASF protocol fault status registers described in *section 3.12*.

3.2 Receive Overflow Errors

If during packet reception, the receive SRAM becomes too full to accept the incoming packet, the packet will be dropped and an interrupt (if not masked) will be generated to inform the system software that packet loss has occurred.

For a correctly configured system with adequate AMBA and software processing bandwidth, such errors should never occur and if packet loss is not acceptable then this interrupt signaling could be used to either adjust system bandwidth to avoid further packet loss or perform some form of recovery as necessary.

Note that the Controller IP itself is capable of handling and recovering from overflow errors due to insufficient system bandwidth. The system integrator should determine if the system bandwidth will always be high enough to prevent overflows from occurring and if occasional packet loss is expected and acceptable from the system design point of view. A safety critical system relying on reliable packet reception should ensure that this error never occurs and transition the system to a safe state for recovery if a fault does occur, potentially indicating a fault within the Controller IP itself or in the overall system.

This error is reported via bit 10 of the generic interrupt status register (separate for each queue) and in bit 20 of the ASF protocol fault status registers described in *section 3.12*.

3.3 Transmit Retry Exceptions

For automotive applications, it is assumed that full duplex operation will always be used. Transmit retry only occurs as a result of collisions in half duplex operation which should never happen in such applications therefore it can be assumed that if such an interrupt is signaled, an unexpected fault must have occurred within the Controller IP and the system should be moved to a safe state and the Controller IP re-initialized.

This error is reported via bit 5 of the generic interrupt status register (separate for each queue) and in bit 8 of the ASF protocol fault status registers described in *section 3.12*.

3.4 Transmit Underruns

For safety critical applications, it is assumed that full store and forward operation would be used for transmitting packets. In this mode, a full Ethernet packet is stored in the transmit SRAM before transmission on the link side interface begins. The underrun error signaling is used to indicate to the system software that the Controller IP has ran out of data while transmitting a packet. This is impossible to occur when operating in this mode and would likely be caused by an unexpected fault in the internal Controller IP signaling and the system should be moved to a safe state and the Controller IP re-initialized.

This error is reported via bit 4 of the generic interrupt status register and in bit 16 of the ASF protocol fault status registers described in *section 3.12*.

3.5 Transmit Data Path Lockup Detection

Various faults that occur within the core could eventually trigger a lockup event where no further packets progress through the transmit data path. A simple lockup detection circuit is included within the IP and monitors for expected packet activity at several points in the data path.

This checks for the following within the DMA part of the design:

- Ensures that each time software issues a 'TX Start' command, a complete packet is stored into the SRAM within '*DMA Lockup Time*'. This check is implemented for each priority queue and is automatically cleared if a 'used bit' is read or a major DMA error occurs.
- For every packet that is written into the SRAM (for each priority queue), a packet count is incremented and is decremented each time a packet is fully transferred to the Controller IP. If there are no packets transferred to the Controller IP within '*DMA Lockup Time*', an error is signaled. The timer portion of this check can be individually enabled or disabled for each priority queue to handle cases where a fixed priority scheme is used and lower priority queues may be temporarily prevented from transmitting. The software may disable the timer for the low priority queues when it is known that high priority transmission must take place.

Note that '*Large Send Offload*' operation is not supported by this mechanism and should not be enabled if LSO is used.

A similar check is implemented for the Controller IP transmit part of the design where:

- For each complete packet transferred to the Controller IP from the DMA, a packet count is incremented and is decremented each time a packet is fully sent on the MII. If there is no activity on the MII after '*MAC TX Lockup Time*', an error is signaled.

The '*DMA Lockup Time*' and '*MAC TX Lockup Time*' values are software programmable and utilize a 'prescale counter' register which is operating in the Controller IP *tx_clk* domain to provide a known time base.

This mechanism is expected to be able to detect many faults and is recommended to be used in conjunction with sections 4.2 and 4.3.1 for independent validation.

This error is reported via the ASF Fault Logging and Reporting module as a transaction timeout error and may be individually masked. Refer to the *User Guide [Table 1]* for more information.

3.6 Receive Data Path Lockup Detection

Various faults that occur within the core could eventually trigger a lockup event where no further packets progress through the receive data path. A simple lockup detection circuit is included within the IP and monitors for expected packet activity at several points in the data path.

This checks for the following within the DMA part of the design:

- For each complete packet transferred to the DMA from the Controller IP, a packet count is incremented and is decremented each time a complete packet has been transferred to system memory. If there are no packets transferred to system memory within '*DMA Lockup Time*' then an error is signaled.

Note that '*Receive Side Coalescing*' is not supported by this mechanism and should not be enabled if RSC is used.

For the Controller IP RX part of the design, a simpler 'receive activity' check is implemented whereby the lockup detect mechanism monitors for a good packet to be transferred to the DMA within '*MAC RX Lockup Time*' and this requires the system to be setup such that periodic packet transfers occur within a known maximum time interval.

The '*DMA Lockup Time*' and '*MAC RX Lockup Time*' values are software programmable and utilize a 'prescale counter' register which is operating in the Controller IP tx_clk domain to provide a known time base.

This mechanism is expected to be able to detect many faults and is recommended to be used in conjunction with sections 4.2 and 4.3.2 for independent validation.

This error is reported via the ASF Fault Logging and Reporting module as a transaction timeout error and may be individually masked. Refer to the *User Guide [Table 1]* for more information.

3.7 Control, Configuration, and Status Register Space Corruption Detection

The IP contains many configuration registers which are expected to be programmed by the system software at start of day and not expected to change during operation. As such, it is possible that a fault could occur on these registers causing unexpected behavior of the Controller IP.

When the design is configured with the '*gem_asf_enable*' Verilog define configuration option, the following mechanisms are added to protect critical regions of the register space:

- All configuration and control registers are parity protected and monitored continuously for parity faults.
- All mission critical status registers (interrupt, transmit and receive status registers) are duplicated and monitored continuously for differences between original and duplicate set.
- Select configuration/control registers that are not well protected by parity (e.g. hardware modifiable registers) are duplicated and protected in the same was the status registers.

If the design is configured to utilize host parity buses, then the stored parity for the configuration/control registers will be from the host interface to provide end to end protection of register writes.

If an error is detected in any of the registers, an interrupt is generated to notify the system software of possible corruption. The Controller IP should then be put into a safe state and re-initialized. System software can then perform register write-read tests to determine if the fault was due to a permanent or transient failure event.

The following registers are not considered to be mission critical and are not protected:

- GPIO
- Statistics

This error is reported via the ASF Fault Logging and Reporting module and may be masked or configured to generate a fatal or non-fatal interrupt. Refer to the *User Guide [Table 1]* for more information.

3.7.1 Diagnostic Coverage (DC)

Parity is stored alongside the written data to detect the following faults:

- Stuck at 0/1 of stored data bits (only cases where the fault differs from intended behavior, e.g. a stuck at 0 fault of a register bit which is written with value of 0 will not cause faulty behavior)
- Transient faults and single bit (element) upsets
- Fault of write/read control logic

To cover the last case of protecting write/read logic, the following use model is assumed:

1. Software writes data to register
2. Hardware stores data with parity
3. Software read back value to ensure register was written correctly
4. Hardware will constantly check the stored data against the stored parity value

This ensures that for the DC calculation, we can focus on subsequent faults of the register.

In the following calculations, we make an assumption on the probability distribution of the number of bits that will be affected by an event:

Probability	Number of Affected Bits
90%	1
9%	2
1%	>2

Note that this distribution determines what the possible outcomes are if a fault event occurs. The probability of the fault event occurring is handled by the underlying calculations within the FMEDA spreadsheet provided by SGS-TÜV and takes into account the 'distribution' column to weight the probability. This means that if a larger bus is considered, this would take on a higher distribution value hence will have more chance of an event happening.

A conservative DC value of 90.5% is used for the FMEDA entry, in reality this figure would be much higher as many of the control and configuration registers are 32-bits wide.

3.7.1.1 1-Bit Parity for 8 Bits of Data

Using the probability distribution above we have the following:

- 1-bit error occurs 90% of the time with detection rate of 100%
- 2-bit errors occurs 9% of the time however this does not change the parity so is undetectable, therefore detection rate is 0%
- 3 or more bit errors occurs 1% of the time. This is only detectable if the number of errors is 3, 5, or 7 bits and is undetectable for 4, 6, 8 bits therefore detection rate is 50%

Combining the detection rate with the probability distribution we get the following DC:

- $0.9 \times 100 + 0.09 \times 0 + 0.01 \times 50$
- **=> DC = 90.5%**

3.7.1.2 2-Bit Parity for 16 Bits of Data

Using the same probability distribution as before, we have the following:

- 1-bit error occurs 90% of the time with detection rate of 100%
- 2-bit errors occurs 9% of the time however this is undetectable if both errors fall within the same byte which can occur in 7/15, therefore detection rate is 53.3%
- 3 or more bit errors occurs 1% of the time. This is always detectable if the number of errors is an odd number OR there is an odd number of errors in either byte. For simplicity we can just consider this as undetectable half the time it is even. Therefore, detection rate is $50\% + 25\% = 75\%$.

Combining the detection rate with the probability distribution we get the following DC:

- $0.9 \times 100 + 0.09 \times 53.3 + 0.01 \times 70$
- **=> DC = 95.5%**

3.7.1.3 4-Bit Parity for 32 Bits of Data

Using the same probability distribution as before, we have the following:

- 1-bit error occurs 90% of the time with detection rate of 100%
- 2-bit errors occurs 9% of the time however this is undetectable if both errors fall within the same byte which can occur in 7/31, therefore detection rate is 77.4%
- 3 or more bit errors occurs 1% of the time. This is always detectable if the number of errors is an odd number OR there is an odd number of errors in any byte. As the number of errors increase, the number of potential undetected faults decrease. For a pessimistic worst case view, we can take the detection rate when it is even to be better than the 2-bit case. This gives an average detection rate of 88.7%

Combining the detection rate with the probability distribution we get the following DC:

- $0.9 \times 100 + 0.09 \times 77.4 + 0.01 \times 88.7$
- **=> DC = 97.9%**

3.8 Data and Address Path Protection

The Controller IP is a very data centric design with large amounts of data buffering and pipelining and some degree for byte level alignment. This sort of structure lends well to using an extra bit per byte parity scheme to detect any single bit corruption on these data path elements which may cause multiple safety goal violations including lockup, frame corruption, and incorrect status reporting.

This is a configuration option and is enabled by setting the '*gem_asf_enable*' Verilog define option as described in the Controller IP *User Guide* [Table 1].

When this option is enabled, key parts of the data path between the host AMBA interface and the GMII interface are protected with an additional 1-bit per byte parity protection scheme such that any single bit fault within a byte can be detected. In addition to the data, the host address buses are similarly protected with a parity scheme to ensure any corruption of the addresses can be detected.

This error is reported via the ASF Fault Logging and Reporting module and may be masked or configured to generate a fatal or non-fatal interrupt. Refer to the *User Guide* [Table 1] for more information.

When a fault occurs, the Controller IP should be transitioned to a safe state and system software may attempt to re-initialize the Controller IP to retry the frames and determine if the fault is permanent or transient.

This mechanism will protect against both permanent and transient fault events affecting the data paths.

3.8.1 Diagnostic Coverage

The diagnostic coverage calculation for the data and address path parity protection is calculated in the same way as in section 3.7.1 and the same probability distribution is used.

As the data path is generally wider at 64-bits or more, the DC increases to 99%. For the purposes of the FMEDA, a conservative value of 98% is used.

3.9 TSU Timer Protection

The TSU timer is used within the core to provide real time indication at the hardware level for use in systems that are using the precision time protocol (PTP) as defined in IEEE STD 1588 or 802.1AS. It is used for many time-based functions within the core. Faults that occur in the timer will have an immediate effect on the control systems that rely on these functions and detection at the system level may take too long. Protection at the hardware level mitigates against that.

If the '*gem_asf_prot_tsu*' Verilog define configuration option is enabled, the TSU is duplicated and there will be two instances of the TSU timer module. The outputs of the modules are compared continuously such that if any fault occurs in either of the TSU units affecting the timer values it will be immediately captured and signaled to the user through the ASF Fault Logging and Reporting module.

In addition to duplication of the TSU module, the timer value is protected throughout the data path with parity protection as defined in *section 3.8*.

If a fault is detected and the TSU operation is necessary for the safe operation of the system, the Controller IP should be transitioned to a safe state and system software may attempt to re-initialize the Controller IP to retry the frames and determine if the fault is permanent or transient.

3.10 Scheduler Protection

A transmit scheduler is used to prioritise traffic from different queues. If a fault occurs in this module, the desired Quality of Service (QoS) parameters may be violated which may not be easily detectable by other means. If accurate QoS is critical for the end application, the scheduler functions may be duplicated to ensure no faults occur in the scheduling of transmit traffic.

This is a configuration option and is enabled by the '*gem_asf_prot_tx_sched*' Verilog define.

When this protection mechanism is enabled, the outputs of the modules are continuously monitored for differences and any deviation reported immediately as a fault through the ASF Fault Logging and Reporting module.

If a fault is detected, it is possible that the communication link may continue to be usable with indeterminate QoS however the Controller IP should be transitioned to a safe state and re-initialized as soon as possible to attempt to restore normal operation.

Note that if a critical failure occurs within the scheduler preventing packets from being transmitted (lockup condition), this will be detected by the TX lockup detection mechanism described in *section 3.5*.

3.11 SRAM Fault Detection and Correction Protection

The DMA portion of the IP uses SRAMs to store transmit and receive data as frames are processed. Faults that occur in the SRAM can cause lockup, frame corruption, incorrect status reporting and unpredictable behavior.

The Controller IP may be configured to add ECC protection (SECDED) to the SRAM data path such that any single bit fault of the SRAM data can be automatically corrected or any two-bit faults may be detected. In both cases, the user is notified through the interrupt notification process.

When this fault occurs, the Controller IP should be transitioned to a safe state and system software may attempt to re-initialize the Controller IP to retry the frames and determine if the fault is permanent or transient.

This protection mechanism is enabled using the '*gem_asf_ecc_sram*' Verilog define option. Refer to the *User Guide [Table 1]* for changes to the SRAM data widths to support ASF options.

3.12 Protocol Fault Checking

The IP can be configured to generate an interrupt based on various built-in checks relating to the line side protocol and general operation of the IP. A mask register is provided to enable the user to select which faults would generate an '*asf_protocol_err*' status.

The following table lists the bit assignments for the various error sources:

Register Information				
Description	"Control register to mask out ASF Protocol faults from triggering interrupts. On reset, all bits are set to mask out all sources. Clear the corresponding bit to enable the interrupt source."			
Offset	0x40		Type:	RW
Bitfield Details				
Bits	Name	Description	Access	Reset
31:22	Reserved	Reserved	RO	0x0
21	rx_dma_pkt_flush	A RX packet was flushed from the DMA	RW	0x1
20	rx_overflow	DMA buffer overflow and packet was dropped	RW	0x1
19	rx_hresp_err	DMA hresp error (AHB only)	RW	0x1
18	tx_hresp_err	DMA hresp error (AHB only)	RW	0x1
17	tx_buff_ex_mid	Transmit buffers exhausted before end of packet.	RW	0x1
16	tx_underrun	Transmit DMA underrun occurred.	RW	0x1
15:9	Reserved	Reserved	RO	0x0
8	tx_too_many_retries	Too many retry attempts after collision (half duplex only)	RW	0x1
7	rx_udp_ck_err	RX packet UDP checksum error	RW	0x1
6	rx_tcp_ck_err	RX packet TCP checksum error	RW	0x1
5	rx_ip_ck_err	RX packet IP checksum error	RW	0x1
4	rx_length_err	RX packet with length field error	RW	0x1
3	rx_symbol_err	RX packet with symbol errors	RW	0x1
2	rx_long_err	RX packet too long	RW	0x1
1	rx_short_err	RX packet too short	RW	0x1
0	rx_crc_err	RX packet with bad CRC	RW	0x1

Some of these error conditions are informational and expected to occur even in a normal system (e.g. CRC errors) and the user should decide which events are relevant based on the system architecture.

4 External Safety Mechanism

The following optional safety mechanisms are recommended to be implemented by the system developer to protect against unexpected faults that may not be detected by the internal safety mechanisms. These faults may be due to physical damage to the hardware or transient faults due to noise, EMI or particles. Implementation of both internal and external diagnostic mechanisms will offer maximum reliability.

Note that implementing or not implementing the safety mechanisms will affect the SPFM/LFM calculation for the IP and will alter the possible ASIL classification level for the IP. If some of the safety mechanisms are not implemented, Cadence will be able to provide the effect estimate.

4.1 Frame Data Validation for End-to-End Data

The following performance enhancing features of the IP may be disabled to provide further data integrity resilience to be performed at a higher layer:

- IP/TCP checksum generation on transmit packets
- IP/TCP checksum validation on receive packets

The IP/TCP checksum generation affects the data path such that at the point of checksum insertion, the internal data path parity defined in *section 3.8* is re-generated. A fault at the checksum generation point may potentially be undetected by the internal data path parity mechanism and a 'good' checksum may be sent out for 'bad' data.

To provide full end-to-end packet integrity checking and guaranteed packet delivery, it is recommended that a high-level protocol such as TCP is used, ensuring that packets are acknowledged or retried as necessary as part of the protocol. In this case by not supporting the hardware TCP checksum offload feature and using the capability within the software stack, full end to end protection can be guaranteed.

It is assumed this detection mechanism will cover both stuck-at-faults and transient faults.

4.2 Periodic Transfer Test

To detect various stuck-at faults, this test should be triggered periodically within the required fault transition time (FTT). The requirement is to have an Ethernet frame transmitted from the local host, and an expected response frame received and captured within the FTT. The method for implementing this can be hardware (packet loopback through external switch) or software (e.g. like PING) and is left to the system integrator to decide on the best approach for their application.

Note that this must be periodic within the FTT to be able to check for stuck-at faults and will not cover all transient faults as the transient event may not coincide with the test packet.

This test should be performed across all configured priority queues.

4.3 Lockup Detection

Lockup detection is a key safety mechanism. It can be implemented externally in software by routinely polling statistics/status/descriptor tables and monitoring for lack of expected activity. It is assumed this detection mechanism will cover both stuck-at-faults and transient faults. An example of how this can be achieved for both transmit and receive is provided below.

Note that the defined mechanism in *section 4.2* may also be used as the basis for this check. There are also pre-defined internal safety mechanisms defined in *section 3.5* and *3.6* which would complement these functions and provide independent checking capability.

4.3.1 Transmit Watchdog

The IP requires a finite time to transmit a given packet. If there are packets scheduled for transmission, a 'transmit complete' interrupt should be generated within a finite time. This watchdog timer should reset each time a 'transmit complete' interrupt is received.

If the watchdog time reaches e.g. 1ms, this may indicate an unknown fault has occurred and the IP should be transitioned to the safe state.

If there are no packets scheduled for transmission, the watchdog timer should be disabled.

The timeout will be application specific and dependent on the speed of operation and the maximum size of the packets being transmitted and worst case bandwidth allocation. For the case where the IP is operating at 100Mbps and the queue bandwidth allocation is set to 1% with no other queue scheduled for transmission, the interrupt should be generated within 500µs. This response time may be further reduced by setting constraints on system maximum packet sizes and minimum bandwidth allocation policies.

4.3.2 Receive Watchdog

A similar liveness check can be implemented for the receive data path. This may rely on the system being set up such that packets are guaranteed periodically such as specified in *section 4.2* as a periodic check to ensure that the test packet can be received.

4.4 Detecting DMA Descriptor Write-Back Address Corruption

In the event of a fault in the DMA control logic, it is possible that incorrect descriptor addresses are generated on the AXI interface.

This corruption can be externally detected by a hardware or software monitor because the IP should always write descriptors to the descriptor table in a sequential manner. If there is an unexpected gap, then it can be considered that a fault has occurred. If there is no update at all to the descriptor table, then the IP can be considered to have locked up and the lockup detection in *section 4.3* should therefore also be used.

To ensure this mechanism can be used, frames should be contained in a single buffer.

This mechanism can be replaced by the internal safety mechanism defined in *section 3.8* and is available in r1p11 and newer revisions of the IP where the internal address paths are protected by parity.

4.5 Detecting DMA Descriptor Write Data Corruption

The descriptors contain both control/data buffer description bits and status bits. Software is responsible for providing all the control/data buffer description information, while the IP hardware is responsible for updating the status bits after frame transmission/reception. To check for descriptor write-back corruption, software should take the following steps:

1. Software should ignore the state of the control/data buffer informational bits (i.e. the bits that the software originally wrote) and never assume those bits are valid for reuse. I.e. always write the control/informational bits for every data buffer being queued and not assume using read-modify-write operation on these bits.
2. For transmit buffer descriptors, none of the status bits should become set under normal operation, other than the TX USED BIT and the timestamp capture bit. If any of the other status bits of the descriptor are set (which may occur as a result of a fault), then this can be monitored and used to force the IP to the safe state. If the timestamp capturing is deemed safety critical (which would be dependent on application the IP is used in, and use cases), then the mechanisms specified in *sections 3.8* and *3.9* should be enabled.
3. For receive buffer descriptors, most of the status bits are informational and are not generally considered as safety critical, e.g. the address match status. However, this is dependent on the use case and the application the IP is used in. The key critical status in the RX descriptor is the length field. If a fault occurs on this field it could be detected through the external data integrity mechanism already specified in *section 4.1*. If the user deems the timestamp capturing status bits as safety critical then the mechanisms specified in *sections 3.8* and *3.9* should also

be enabled. If the user deems any of the other status bits are safety critical then they can be verified in software by implementing a duplicate status generation function external to the IP and comparing the result.

The internal safety mechanism defined in *section 3.8* also provides parity protection for the status buses as they are buffered and pipelined throughout the design and is expected to be able to cover most potential faults.

4.6 Detecting Incorrect Status or Interrupt Generation

Customers should determine which of the events in the TX/RX status and/or interrupt status registers are deemed as fatal. Some guidance on this and how to detect faults in each event is provided below.

- `tsu_timer_comparison_interrupt` - possibly safety critical depending on use case. Detectable by implementing similar function in SW or external HW. The function itself is explained in the IP *User Guide [Table 1]* and not repeated here. Note that the time value itself is parity protected allowing this function to be reliably implemented externally if necessary.
- `wol_interrupt` – this is not assumed to be used in automotive applications.
- Receive LPI indication - Possibly safety critical depending on use case. Assuming LPI is supported and the connected PHY is able to report LPI state, then SW can validate the LPI state by reading both PHY and MAC status registers. Otherwise, LPI should not be used.
- TSU seconds increment - possibly safety critical depending on use case. Detectable by implementing similar function in HW by interrogating the timer output which is parity protected.
- 1588 Frame decodes – these are informational only and the timestamp information within the descriptors are generally used.
- PCS Interrupts and status: assumed PCS not present in automotive applications.
- External Interrupt Input – this is not fault protected and should not be used for safety critical applications.
- 802.3/PFC Pause frame interrupts - assumed not safety critical and these interrupts are generally considered informational.
- AXI Response error or AMBA error - Should not receive these at any time. If they are triggered either correctly or as a result of a fault, enter safe state immediately.
- Receive Overrun – if this is triggered, it indicates a possible issue with the system bandwidth setup. The IP cleanly handles dropping receive frames due to insufficient storage in the receive SRAM. If a fault causes this interrupt to not trigger correctly, then the missing packet would be detectable through one of the previously defined lockup or packet integrity check mechanisms.
- Transmit/Receive Complete - Safety critical. This is detectable by using the mechanisms in **Error! Reference source not found.** and **Error! Reference source not found.** to detect failure of these interrupts. If these interrupts are inadvertently triggered, software will as part of normal operation check the descriptor tables and not take any further action if no new frame is seen.
- `retry_limit_exceeded_or_late_collision` - half duplex assumed not to be enabled, so should not occur. If these are seen, this indicates a possible fault in the transmitter and the IP should be placed into safe state.
- underrun - should not occur as design is assumed to use full store and forward in safety critical applications. If they are triggered as a result of a fault, enter safe state immediately.
- Descriptor Used Bits Read (TX) - not deemed safety critical as they are informational.
- Descriptor Used Bits Read (RX) - not deemed safety critical as they are informational.
- `management_frame_sent` (MDIO transfer complete) - possibly safety critical depending on use case. MDIO transfers are triggered by software and can be managed by interrupts or status polling. If this interrupt is not triggered or status not set within 500 MDC cycles, then it is considered to have failed. The MDIO transfer should then be repeated. If it fails a second time, this may indicate a failure to communicate with the PHY, move to the safe state. Ignore the triggering of this interrupt when it is received at a non-expected time.

Other status reported via APB reads are assumed not safety critical. It is assumed this detection mechanism will cover both stuck-at-faults and transient faults.

The internal safety mechanism defined in *section 3.7* is expected to protect against any faults within the registers modules but may not protect against all faults originating at the interrupt/status source.

4.7 Validation of APB Accessible Configuration Registers

To validate the state of the writable configuration/control registers of the IP, a periodic compare of these registers against a stored mirror copy in software can be implemented. This should be periodic within the FTT to be able to check for faults.

When performing configuration of the IP, the registers should be read back to ensure proper capture of the configuration as expected.

This can be used in conjunction with the internal safety mechanism in *section 3.7* to limit the registers to be validated to the dynamic control registers used as part of normal frame handling operations.

4.8 Interrupt Pin Validation

SW can write to the interrupt status register to force the interrupt pin to trigger. This can be done periodically to check the interrupt pin is operational. This should be periodic within the FTT to be able to check for faults.

As part of *section 4.2*, key interrupt and status pins will also be utilized periodically.

4.9 Redundancy

For applications where the safe state of the IP is not acceptable, e.g. situations where shut down of the communication channel violates the system safety goals, a possible solution is to have multiple communication paths using multiple IP instances and software generation and discarding of duplicate packet data. This may protect against random faults caused by EMI etc. Note this mechanism is not required to protect against faults that violate the specified SGV's in *section 2.1*, but is described as a backup or as a potential alternative to the mechanisms stated above.

Change Log

Version	Description of Change
0.1	April 29, 2016 Initial draft version.
0.2	May 13, 2016 Updated acronyms, base failure rate section, safety mechanisms and safety goals. Updated front title page. Add comment about availability of external TSU option and suggest not using multi-buffer transmit operation for external safety mechanism section.
0.3	June 8, 2016 Added further information for internal safety mechanisms. Added status column in change history. Update document and part numbers
0.4	July 12, 2016 Added FMEDA reference only
0.5	Nov 30, 2016 Updated safety mechanism description to reflect revision 1p10 of the IP.
0.6	Dec 2, 2016 Minor comments added to equation explanation and safety mechanism descriptions.
0.7	April 5, 2017 Initial update for revision 1p11 of the IP.
0.8	June 1, 2017 Updated descriptions for safety features and FIT rate to match new IP.
1.0	June 16, 2017 Editorial review – aligned style to new template.
1.1	September 22, 2017 Update to revision 1p12 of the IP.